This question paper contains 4+1 printed pages + 1 Attached Chart]

Roll No. | | | | | | | | | | | |

S. No. of Question Paper : **2657**

Unique Paper Code : **32517916** **IC**

Name of the Paper : **Embedded Systems**

Name of the Course : **B.Sc. (Hons.) Electronics : DSE-3**

Semester : **VI**

Duration : **3 Hours** Maximum Marks : **75**

(Write your Roll No. on the top immediately on receipt of this question paper.)

Question No. **1** is compulsory.

Attempt any *four* questions out of the remaining six.

All questions carry equal marks.

Write answers of all parts of a question at one place only.

1. (*a*) What is the function of brownout detector ?

(*b*) What is the function of "SP Register" ? What is the size of SP register in ATMEGA32 ?

(*c*) Differentiate between the following :

(*i*) LSR and ASR

(*ii*) SBI and SBIC.

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(*d*) Find the contents of register after execution :

PORTB = ~ (1<<2)

PORTC = (0×34) & (1<<3)

(*e*) Give *two* advantages of SPI devices. 3×5=15

2. (*a*) What is an embedded system ? What are its essential features ?

(*b*) Briefly explain the Data memory map of ATMEGA32.

(*c*) Explain the difference between RISC and CISC processors. 5+5+5

3. (*a*) Explain the function of the following instructions with an example :

(*i*) BRSH

(*ii*) SBIW

(*iii*) LD

(*iv*) PUSH

(*v*) STS.

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(b) What is the value of R20 and carry flag after the execution of the following code :

```
LDI R20, 0 × 40
CLC
ROR R20
ROR R20
ROR R20
ROR R20
```

(c) Compare and contrast C programming language over assembly language. 5+5+5

4. (a) A door sensor is connected to the PORT B pin 1, and an LED is connected to PORT C pin 7. Write an AVR C/assembly program to monitor the door sensor and, when it opens, turn on the LED (when door opens, state of PB1 is logic '1').

(b) Write a program to monitor PB7, if it is at logic '1' make PB4 input else make PB4 as output.

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(c) Write a program to display first ten terms of Fibonacci series at Port B. 5+5+5

5. (a) What is the difference between 'Interrupt' and 'Polling' ?

(b) What are the various sources of interrupt in AVR ? Specify the triggering options of hardware interrupts.

(c) Write a main program to initialize the interrupt and enable INT0. Write an interrupt service routine that increments the contents of R20 whenever INT0 is triggered. 5+5+5

6. (a) Explain with a diagram to show how Timer can be used as delay timer and event counter.

(b) Draw diagrams of all four modes of timer.

(c) An LED is connected to PB4 pin of Port B. Assuming system clock frequency XTAL = 8MHz, write a program that toggles LED whenever TOV0 flag is set. 5+5+5

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7. (*a*) For a 10-bit ADC :

(*i*) Determine the output voltage if reference voltage V_{ref} is 5 V and analog input voltage is 0.488 V.

(*ii*) What will be the output if V_{ref} = 2.56 V and analog input voltage is 0.2 V ?

(*b*) Explain serial communication through I2C bus protocol.

(*c*) Compare and contrast SPI protocol with I2C protocol. 5+5+5

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